TWO-STAGE STUDENT HACKATHON PREPARATION GUIDE

Stage 1: Problem Framing | Stage 2: MRD (Market Requirements Document)

Purpose: To guide students from identifying and validating a real-world problem to defining a practical, measurable and prototype-ready solution.

Stage	Focus	Key Question	Main Student Output
Stage 1	Problem Framing	What exactly is the problem, who faces it, and why does it matter?	Problem Framing Report + Problem Statement
Stage 2	MRD	What should the proposed solution provide, for whom, and with what measurable requirements?	MRD + Solution Requirements + MVP Plan

Core Principle

Problem First → **Solution Next** → **Technology Last.** Students should not begin by selecting a technology and then searching for a problem. They should first establish the user need and evidence.

STAGE 1 – PROBLEM FRAMING

Objective

Students identify, understand and validate a real-world problem before proposing a technical solution.

1. Problem Identification

Identify problems from college/campus, hostel, transportation, agriculture, healthcare, education, environment, manufacturing, banking, retail, government services, rural communities, industry, smart city, cybersecurity, AI/data science and other domains.

2. Problem Observation

Observe the actual situation. Record where, when and how the problem occurs instead of starting with an application or technology idea.

Example: “Hostel students spend considerable time waiting in queues during peak mess hours.” This is a problem. “We want to develop an AI-based mess application” is already a solution concept.

3. Identify Target Users

- Who experiences the problem?
- How many users are affected?
- Where does it occur?
- When and how frequently does it occur?
- What method is currently used?

4. Define Pain Points

Pain Point	Possible Impact
Long queue	Students waste time
No crowd visibility	Students cannot choose an appropriate time
Peak-hour congestion	Mess becomes overcrowded
Difficult quantity estimation	Food wastage or shortage
No historical data	Management cannot predict demand

5. Root Cause – 5 Why Analysis

Example: Problem: Students wait too long at the mess → too many students arrive together → students do not know the crowd level → no real-time crowd information → no monitoring/prediction system → lack of data-driven demand management.

6. Collect Evidence

- Student/user interviews
- Google Forms survey
- Direct observation
- Photographs or process records
- Institutional/industry data
- Research papers and credible reports

- Expert or stakeholder interviews

Suggested validation: 10–20 user responses for a mini-hackathon; 50–100 responses for a larger hackathon, where practical.

7. User Interview Questions

- How often do you face this problem?
- What difficulties do you experience?
- How do you currently solve it?
- How much time or money is lost?
- What is the biggest pain point?
- What improvement would you expect?
- Would you use a digital solution?
- Which features would be useful?

8. Problem Validation

- Is the problem real and frequent?
- Is it significant to users?
- Are the affected users clearly identifiable?
- Are current solutions inadequate?
- Is the problem technically solvable?
- Is it suitable for a hackathon?

9. Final Problem Statement

Recommended format: “[Target users] face [specific problem] in [specific context], resulting in [impact]. Existing methods such as [current approach] are insufficient because [gap]. Therefore, there is a need for [desired outcome].”

STAGE 1 – STUDENT SUBMISSION

Problem Framing PPT – 5 to 7 Slides

Slide	Required Content
1	Problem title, team name, members, department
2	Problem background and current situation
3	Target users and stakeholders
4	Pain points and impact
5	5-Why/root-cause analysis
6	Survey, interview and observation evidence
7	Final validated problem statement

Stage 1 Evaluation – 100 Marks

Criteria	Marks
Problem clarity	20
User identification	15
Pain-point analysis	15
Root-cause analysis	15
Evidence / survey	15
Problem validation	10
Final problem statement	10
Total	100

Stage 1 Gate

Only teams with a clearly defined, evidence-supported and validated problem should proceed to Stage 2.

STAGE 2 – MRD

MRD = Market Requirements Document

For a student hackathon, the MRD defines the target users/market, their needs, required solution capabilities, priorities, constraints and measurable success criteria.

1. Executive Summary

Summarize the problem, target users, proposed solution, major benefit and expected impact.

2. Target Market / User Segment

Category	Example
Primary users	Hostel students
Secondary users	Mess staff
Decision maker	Hostel administration
Beneficiary	Institution
Potential market	Colleges / hostels

3. User Persona

Student Persona: age/role, current problem, goal, technology usage and expectations. **Administrator Persona:** operational goal, current difficulty, required information and decision-making needs.

4. User Journey

Current: Student finishes class → goes to hostel → checks mess → finds large crowd → waits → loses time.
Proposed: Student opens system → views current crowd → checks predicted rush → chooses suitable time → visits mess → reduced waiting.

5. Solution Concept

Describe the solution at a high level. Example modules: student application, real-time crowd information, prediction engine, notifications, administration dashboard, demand analytics.

6. Functional Requirements

ID	Requirement	Priority
FR01	User login	Must Have
FR02	Display current crowd	Must Have
FR03	Predict peak period	Must Have
FR04	Recommend suitable time	Must Have
FR05	Admin dashboard	Must Have
FR06	Meal demand prediction	Should Have
FR07	Notifications	Should Have
FR08	Historical analytics	Could Have

7. Non-Functional Requirements

- Performance

- Security and privacy
- Scalability
- Availability
- Usability
- Reliability
- Response time

MRD – DETAILED REQUIREMENTS

8. Technology Requirements

Select technologies only after understanding the problem. Possible choices include Flutter/React Native, web applications, Python/Node.js/Java, MySQL/PostgreSQL/Firebase, ML models, IoT sensors, camera-based counting or QR-based systems.

9. Competitor / Existing Solution Analysis

Existing Approach	Typical Limitation
Fixed timings	No real-time crowd information
Manual observation	Not scalable
Messaging groups	Limited real-time analytics
Manual food estimation	May not accurately predict demand

Students should identify the solution gap instead of simply claiming that their idea is unique.

10. Innovation

Clearly state what is new or improved: AI prediction, IoT integration, automation, new workflow, cost reduction, accessibility, sustainability or improved user experience.

11. MVP – Minimum Viable Product

Define only the essential features that can be demonstrated during the hackathon. Example: login, current crowd display, crowd prediction, recommended timing and admin dashboard.

12. Success Metrics / KPIs

KPI	Illustrative Target
Waiting time	Reduce by 30%
Prediction accuracy	>80%
User adoption	>70%
Food wastage	Reduce by 15%
Dashboard response	< 3 seconds

Targets are proposed goals for the prototype; they should be validated during testing.

13. Business / Sustainability Model

Consider college subscription, institutional licensing, SaaS, government implementation, industry partnership or other appropriate models.

14. Risk Analysis

Risk	Possible Mitigation
Low student adoption	Simple UI and awareness
Incorrect prediction	Continuous model validation/training
Sensor failure	Backup/manual entry
Data privacy	Secure authentication and limited data collection

Risk	Possible Mitigation
Internet failure	Caching/offline fallback where feasible

STAGE 2 – STUDENT SUBMISSION & EVALUATION

MRD Deliverables

- MRD document: approximately 8–12 pages
- MRD PPT: approximately 8–10 slides
- User persona
- User journey
- Solution architecture
- Feature prioritization
- MVP definition
- Technology stack
- KPI / impact measures
- Business or scaling plan
- Risk analysis

MRD Evaluation – 100 Marks

Criteria	Marks
User / market understanding	15
Solution relevance	15
Requirements definition	15
Innovation	15
MVP definition	10
Technical feasibility	10
KPI / measurable impact	10
Scalability / sustainability	5
Presentation clarity	5
Total	100

Two-Stage Hackathon Workflow

REAL-WORLD OBSERVATION → PROBLEM IDENTIFICATION → USER IDENTIFICATION → PAIN POINTS → 5-WHY ROOT CAUSE → SURVEY / INTERVIEW → PROBLEM VALIDATION → FINAL PROBLEM STATEMENT → STAGE 1 REVIEW → SHORTLIST → MRD → TARGET MARKET → USER PERSONA → USER JOURNEY → SOLUTION CONCEPT → REQUIREMENTS → MVP → TECHNOLOGY → INNOVATION → KPI → SCALE → RISK → STAGE 2 REVIEW → PROTOTYPE / PITCH

Final Checklist for Students

- Do not start with technology; start with the user problem.
- Use evidence rather than assumptions.
- Separate problem, root cause and solution.

- Keep the MVP realistic for the hackathon duration.
- Prioritize Must Have features.
- Define measurable KPIs.
- Explain how the solution can scale.
- Be prepared to demonstrate the prototype and defend the assumptions.

Recommended Hackathon Principle: Problem First – Solution Next – Technology Last.